Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 44915

Bachelor of Computer Application

(Semester Third) (Main/Back)

Examination, 2025-26

BCA-205N

Operating System

Time : Two Hours] *[Maximum Marks : 75*

Note : Attempt **all** the Sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt all **ten** questions. Each very short questions carries **2** marks. 2×10=20

1. (a) What is an operating system? Give two examples.

   (b) Define process. How is it different from a program?

https://www.pyqonline.com

(c) What is a system call? Give any two examples.

(d) Define CPU scheduling. Name any two scheduling algorithms.

(e) Define multiprogramming.

(f) What is a context switch?

(g) What is thrashing in an operating system?

(h) Define virtual memory. Why is it used?

(i) What is the purpose of a file allocation table (FAT)?

(j) Differentiate between primary memory and secondary memory.

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions. Short questions carries **15** marks each. Short answer are required not exceeding **200** words.2×15=30

https://www.pyqonline.com

2. Explain the concept of a Process Control Block (PCB).
3. What is the difference between paging and segmentation?
4. Define critical section and explain why it is important in concurrency.
5. What is a page fault? How does an operating system handle it?

Section-C

(Long Answer Type Questions)

Note : Attempt any **one** question. Long questions carries **25** marks each. Long answer are required not exceeding **800** words.

1×25=25

6. Explain various memory management techniques used in operating systems including contiguous allocation, paging, segmentation, fragmentation, and page replacement algorithms. Provide examples.

https://www.pyqonline.com

7. Describe the process life cycle in detail. Explain each state such as new, ready, running, waiting and terminated. Also discuss state transitions with a neat diagram.
8. What is a deadlock? Explain deadlock conditions, prevention, avoidance (Banker's algorithm), detection and recovery methods. Provide a real-world example.

https://www.pyqonline.com